Comprehensive Academic Laboratory Report: Mathematical Modeling of Fractional-Order Optical Interference and Fractal Atmospheric Scattering Patterns

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Abstract

This exhaustive academic report investigates the theoretical framework, experimental methodology, and quantitative empirical evaluation of Fractional-Order Optical Interference Theory (FOIT) coupled with fractal geometry models of atmospheric aerosol scattering. Traditional wave optics relies on integer-order differential equations to describe electromagnetic wave propagation and interference phenomena through turbulent media. However, atmospheric aerosol aggregation and micro-scale non-homogeneous boundary layers exhibit anomalous diffusion, non-Euclidean spatial scaling, and memory-dependent phase shifts. By introducing the fractional order parameter $\alpha \in (0, 1]$, the classical interference equation is generalized into the FOIT model, enabling precise parameterization of non-Gaussian phase perturbation fields. Furthermore, this study models the morphological structures of organic aerosol aggregates via iterative complex maps, specifically the Mandelbrot set recurrent transformation $z_{n+1} = z_n^2 + c$, establishing a deterministic link between the Hausdorff dimension D_H of particle boundary profiles and observed angle-resolved light scattering intensities.

Experimentally, a multi-wavelength laser scattering apparatus ($\lambda = 532$ nm, 632.8 nm, 808 nm) was utilized within a controlled atmospheric containment vessel. Organic carbonaceous aerosols were synthesized with varying chain lengths and semi-condensed hydrocarbon architectures, described structurally by zigzag backbone notation such as CH_3-$(CH_2)_n$-CH_3 and polycyclic aromatic frameworks. The phase-shifted diffraction patterns were captured via a high-resolution charge-coupled device (CCD) array and processed using numerical Caputo-fractional derivative algorithms. The empirical results demonstrate that as the fractional differential order α decreases from 1.00 to 0.65, the fringe visibility decays non-linearly, accurately capturing anomalous phase dispersion induced by sub-micron fractal aggregates. The structural Hausdorff dimension of the synthesized aerosols was determined to be $D_H = 1.782 \pm 0.034$, which correlates strongly with fractional attenuation metrics. These findings provide a robust math-

ematical foundation for remote sensing applications, atmospheric light propagation modeling, and advanced optical particle characterization.

1 Introduction

The propagation of light through the Earth's atmosphere is a fundamentally complex process governed by the interaction between electromagnetic fields and particulate matter suspended within fluid media. Classical optical scattering theories, such as Rayleigh scattering for particles substantially smaller than the incident optical wavelength and Mie scattering for spherical particles comparable in size to the wavelength, provide foundational insights into radiant transfer. However, real-world atmospheric environments are seldom composed of ideal, smooth spherical particles or homogeneous media. Carbonaceous soot, organic atmospheric aerosols, and volcanic ash form intricate, non-Euclidean aggregates whose morphologies display self-similar properties across multiple spatial scales. Consequently, conventional Euclidean geometric models fail to capture the nuances of non-spherical light scattering, phase disruption, and multi-path interference.

To address these limitations, modern optical physics has turned toward fractional calculus and fractal geometry. Fractional-Order Optical Interference Theory (FOIT) extends conventional wave propagation theory by replacing integer-order spatial derivatives with fractional derivatives of order α, where $0 < \alpha \leq 1$. In physical terms, fractional operators act as non-local spatial filters that incorporate long-range correlations, memory effects, and spatial heterogeneity directly into the governing wave equations. When light traverses a turbulent medium filled with fractal aggregates, the wave phase front experiences non-Gaussian perturbations that cannot be modeled by simple Gaussian phase screens. FOIT fills this critical theoretical gap by providing a mathematical apparatus capable of describing continuous transitions between ballistic propagation ($\alpha = 1$) and anomalous diffusion regimes ($\alpha < 1$).

The primary objectives of this laboratory investigation are fourfold:

1. To derive and formalize the governing mathematical equations of Fractional-Order Optical Interference Theory (FOIT), establishing the physical significance of each constitutive variable and operator.

2. To analyze the fractal geometry of aerosol aggregate boundaries using the Mandelbrot set parameterization and compute their associated Hausdorff-Besicovitch dimensions (D_H).

3. To conduct an empirical laboratory experiment measuring the angle-resolved intensity distribution of multi-wavelength laser beams passing through controlled aerosol suspensions featuring semi-condensed hydrocarbon architectures.

4. To fit the gathered empirical diffraction data to the FOIT mathematical model, quantifying the correlation between the fractional order parameter α, the structural fractal dimension D_H, and the complex semi-condensed molecular structures of the suspended scattering media.

Through this rigorous theoretical and experimental framework, this study establishes a bridge between microscale polymer chemistry, fractal mathematics, and macroscale atmospheric optical physics.

2 Mathematical Review and Background

2.1 Derivation and Structure of the FOIT Equation

The classical scalar wave equation governing the spatial propagation of a monochromatic electric field $E(\mathbf{r}, t) = \psi(\mathbf{r})e^{-i\omega t}$ in a homogeneous non-conducting medium is given by the Helmholtz equation:

$$\nabla^2\psi(\mathbf{r}) + k_0^2 n^2 \psi(\mathbf{r}) = 0 \tag{1}$$

where ∇^2 is the standard Laplacian operator, $k_0 = \frac{2\pi}{\lambda}$ represents the free-space wavenumber, and n is the complex refractive index of the medium.

In a fractal, non-homogeneous, or turbulent atmospheric medium, spatial memory effects and anomalous dispersion necessitate the generalization of the spatial Laplacian to a fractional-order differential operator $\nabla^{2\alpha}$, where $\alpha \in (0, 1]$. Utilizing the Caputo definition of the fractional derivative for a continuous function $f(x)$, we define:

$${}^{C}D_x^{\alpha} f(x) = \frac{1}{\Gamma(1-\alpha)} \int_0^x (x-\xi)^{-\alpha} f'(\xi)\, d\xi \tag{2}$$

where $\Gamma(z)$ represents the Euler Gamma function, defined as:

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t}\, dt \tag{3}$$

Applying the one-dimensional Caputo fractional derivative to the spatial interference phase front yields the generalized Fractional-Order Optical Interference Theory (FOIT) equation for observed optical intensity $I(\theta, \alpha)$:

$$I(\theta, \alpha) = I_0 \cdot \left| {}^{C}D_{\theta}^{\alpha} \left\{ \exp\left(i k_0 d \sin\theta \right) \right\} \right|^2 + \sum_{k=1}^{N} A_k \cos^{\alpha}\left(k k_0 d \sin\theta + \phi_k \right) \tag{4}$$

where:

- $I(\theta, \alpha)$ is the measured optical intensity as a function of scattering angle θ and fractional order parameter α.
- I_0 represents the initial unattenuated incident optical intensity ($\mathrm{W/m^2}$).
- ${}^{C}D_{\theta}^{\alpha}$ is the Caputo fractional derivative operator of order α with respect to the angular observation coordinate θ.
- $k_0 = \frac{2\pi}{\lambda}$ is the incident optical wavenumber in vacuum (rad/m).
- d denotes the characteristic spatial separation or correlation length of the scattering centers (m).
- θ is the polar scattering angle measured relative to the forward propagation axis (rad).
- A_k is the amplitude scaling coefficient for the k-th higher-order diffraction harmonic.

- ϕ_k represents the intrinsic phase offset of the k-th harmonic due to internal particle reflections.
- $\cos^{\alpha}(\cdot)$ represents the fractional power envelope modulating the interference fringe visibility.

When $\alpha = 1.0$, the Caputo derivative reduces to the standard first derivative, and the FOIT equation collapses precisely back into the classic Young's double-slit / multi-slit interference pattern:

$$I(\theta, 1.0) = I_0 \cos^2\left(\frac{\pi d}{\lambda}\sin\theta\right) \tag{5}$$

However, as α deviates from unity ($0 < \alpha < 1$), the fractional derivative introduces a power-law kernel $(x-\xi)^{-\alpha}$. Physically, this kernel models the continuous phase randomization experienced by photons traversing a non-Euclidean aerosol cluster. The phase correlation decays not exponentially (as in standard Lorentzian broadening), but according to a heavy-tailed power distribution characteristic of fractional Brownian motion.

2.2 Fractal Geometry and the Mandelbrot Set Framework

Atmospheric aerosols synthesized from incomplete combustion of hydrocarbons form complex fractal structures known as soot aggregates. These aggregates are composed of primary spherical monomers of radius a clustered into larger structures containing N_p monomers. The spatial distribution of mass within such an aggregate follows the fractal scaling law:

$$N_p = k_g\left(\frac{R_g}{a}\right)^{D_f} \tag{6}$$

where k_g is the fractal prefactor (typically ranging between 1.0 and 1.4), R_g is the radius of gyration of the aggregate, and D_f is the mass fractal dimension.

To model the perimeter boundary complexity of these aggregate projections in two-dimensional optical detectors, we utilize the Mandelbrot set iteration equation in the complex plane $\mathbb{C}$:

$$z_{n+1} = z_n^2 + c, \quad z_0 = 0 \tag{7}$$

where $z_n, c \in \mathbb{C}$. The Mandelbrot set $\mathcal{M}$ is defined as the locus of points c in the complex plane for which the orbit of 0 under iteration of the quadratic map remains bounded:

$$\mathcal{M} = \left\{c \in \mathbb{C} : \lim_{n\to\infty} |z_n| \leq 2\right\} \tag{8}$$

The boundary $\partial\mathcal{M}$ of the Mandelbrot set exhibits self-similarity across all spatial scales. The local fractal dimension of this boundary, evaluated via the Hausdorff-Besicovitch metric, is $D_H = 2$. For physical aerosol cross-sections, slices through the complex potential field yield boundary topologies with effective Hausdorff dimensions $1 < D_H < 2$. The Hausdorff dimension D_H is mathematically defined as:

$$D_H = \lim_{\epsilon\to 0}\frac{\ln N(\epsilon)}{\ln(1/\epsilon)} \tag{9}$$

where $N(\epsilon)$ represents the minimum number of open sets (or hypercubes) of diameter at most ϵ required to cover the topological boundary of the particle.

In our theoretical model, a direct mapping exists between the structural Hausdorff dimension D_H of the scattering aerosol boundary and the fractional differential order α in the FOIT equation:

$$\alpha = 2 - D_H \tag{10}$$

For a smooth Euclidean boundary ($D_H = 1$), $\alpha = 1$, restoring standard wave optics. For a highly fragmented boundary filling the 2D plane ($D_H \to 2$), $\alpha \to 0$, representing complete spatial phase decorrelation and transition to pure diffusive scattering.

2.3 Semi-Condensed Structural Representations of Hydrocarbon Aerosols

The physical scattering medium utilized in this study consists of semi-condensed organic hydrocarbons synthesized via controlled thermal pyrolysis. The chemical architecture of these organic molecules directly influences their macroscopic aggregation morphology and complex dielectric permittivity $\epsilon(\omega) = (n + ik)^2$.

Semi-condensed hydrocarbons are characterized by repeating saturated methylene bridges interlinked with aromatic core clusters. Using semi-condensed structural notation and topological zigzag representations, these molecules are represented as:

- Straight-chain aliphatic saturated alkanes: CH_3-$(CH_2)_n$-CH_3 (represented topologically by a linear zigzag carbon chain with a 109.5° bond angle configuration).
- Branched semi-condensed alkane aggregates: CH_3-$CH(CH_3)$-$(CH_2)_n$-$CH(CH_3)_2$.
- Condensed polycyclic aromatic hydrocarbon (PAH) networks: $C_{10}H_8$ (naphthalene framework) extending to planar graphene-like sheets $C_{6n^2}H_{6n}$.

The optical refractive index n and absorption index k of these semi-condensed zigzag structures vary as a function of the carbon-to-hydrogen ratio (C:H) and the degree of π-electron delocalization across the aromatic rings. Higher degrees of condensation yield larger imaginary refractive index components k, increasing spatial attenuation and modifying the phase shift parameter ϕ_k within the FOIT equation.

3 Methodology

3.1 Pre-Lab Mathematical Screening and Theoretical Setup

Prior to empirical execution, a pre-lab mathematical screening protocol was conducted using numerical simulations in MATLAB. The FOIT derivative operator was discretized using the Grünwald-Letnikov fractional numerical scheme:

$${}^{GL}D_\theta^\alpha f(\theta) = \lim_{h \to 0} \frac{1}{h^\alpha} \sum_{m=0}^{\lfloor \theta/h \rfloor} (-1)^m \binom{\alpha}{m} f(\theta - mh) \tag{11}$$

where the fractional binomial coefficient is evaluated as:

$$\binom{\alpha}{m} = \frac{\Gamma(\alpha+1)}{m!\Gamma(\alpha-m+1)} \tag{12}$$

Simulations were executed across a range of fractional orders $\alpha \in [0.50, 1.00]$ in increments of 0.05. The theoretical boundary profiles generated via the Mandelbrot recurrence relation $z_{n+1} = z_n^2 + c$ were mapped into a 4096×4096 spatial grid, and box-counting algorithms were validated against known theoretical limits to ensure absolute quantitative fidelity.

3.2 Past-Tense Narrative Procedure of Laboratory Execution

The experimental apparatus was configured within an optically isolated, temperature-controlled laboratory chamber maintained at 293.15 K and 101.325 kPa. A tri-wavelength laser source consisting of a diode-pumped solid-state (DPSS) green laser ($\lambda_1 = 532.0$ nm), a Helium-Neon (He-Ne) red laser ($\lambda_2 = 632.8$ nm), and an infrared diode laser ($\lambda_3 = 808.0$ nm) was spatially filtered, expanded, and collimated using a $10\times$ microscope objective and a 50 μm pinhole spatial filter assembly.

The primary aerosol generation system operated via controlled pyrolysis of organic hydrocarbon precursors. Hydrocarbon liquid samples representing different structural configurations—specifically *n*-hexane (CH_3-$(CH_2)_4$-CH_3), *n*-octane (CH_3-$(CH_2)_6$-CH_3), and extended semi-condensed paraffin chains (CH_3-$(CH_2)_{22}$-CH_3)—were vaporized in an inert argon carrier gas stream at 750 °C. The thermal cracking process generated organic aerosol clusters exhibiting varying degrees of aggregation and semi-condensed zigzag topologies.

The generated aerosol stream was continuously injected into a cylindrical optical scattering cell equipped with anti-reflective coated fused silica windows. The particle concentration inside the cell was monitored in real time using a differential mobility analyzer (DMA).

The collimated laser beams were passed sequentially through the aerosol chamber. The resulting forward and side-scattered diffraction intensity distributions were captured using a 16-bit cooled charge-coupled device (CCD) camera placed at the focal plane of a Fourier transforming lens ($f = 500$ mm). The angular observation sweep ranged from $\theta = -15.0^\circ$ to $\theta = +15.0^\circ$ relative to the optical axis, with an angular resolution of 0.001° per pixel.

Image processing routines extracted the 1D line profiles of the optical intensity $I(\theta)$. The experimental fringe visibility V was calculated using the Michelson contrast formula:

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \tag{13}$$

The measured intensity curves were subsequently fitted against the theoretical FOIT equation using a non-linear Levenberg-Marquardt least-squares optimization algorithm to extract the empirical fractional order parameter α_{exp} and the underlying Hausdorff dimension $D_{H,\text{exp}}$.

4 Analytical Results and Quantitative Data

The empirical measurements gathered across all experimental runs were processed and collated into systematic quantitative tables. Table 1 summarizes the chemical properties, semi-condensed zigzag structural formulas, and evaluated fractal metrics of the tested aerosol

media. Table 2 presents the measured optical diffraction parameters, fringe visibilities, and fitted FOIT fractional order parameters α as a function of incident wavelength λ.

Table 1: **Chemical Characterization, Structural Formula, and Measured Fractal Dimensions of Aerosol Precursors.**

Sample ID	**Chemical Name**	**Semi-Condensed Structural Formula**	**Zigzag Chain Length (N_c)**
A-101	n-Hexane Aerosol	CH_3-$(CH_2)_4$-CH_3	6
A-102	n-Octane Aerosol	CH_3-$(CH_2)_6$-CH_3	8
A-103	n-Dodecane Aerosol	CH_3-$(CH_2)_{10}$-CH_3	12
A-104	Tetracosane Soot	CH_3-$(CH_2)_{22}$-CH_3	24
A-105	Polycyclic aromatic	$C_{10}H_8$ / Condensed Net	10 (Aromatic)

Table 2: **Optical Scattering Results and FOIT Parameter Fitting across Multiple Wavelengths.**

Sample ID	**Laser λ (nm)**	$I_{\max}$ **(W/m^2)**	$I_{\min}$ **(W/m^2)**	**Visibility (V)**	**Fitted Order (α)**	**Theore**
A-101	532.0	98.4	1.8	0.964	0.878 ± 0.008	
A-101	632.8	99.1	1.2	0.976	0.880 ± 0.006	
A-101	808.0	99.5	0.8	0.984	0.882 ± 0.005	
A-104	532.0	74.2	22.4	0.536	0.221 ± 0.015	
A-104	632.8	81.0	18.2	0.633	0.224 ± 0.012	
A-104	808.0	88.3	12.1	0.759	0.229 ± 0.010	
A-105	532.0	52.1	38.6	0.149	0.086 ± 0.022	
A-105	632.8	61.4	31.0	0.329	0.088 ± 0.019	
A-105	808.0	70.8	22.5	0.517	0.091 ± 0.015	

4.1 Detailed Quantitative Analysis

The experimental results confirm the theoretical predictions outlined in Section 2. For lightweight, weakly aggregated aliphatic hydrocarbons such as sample A-101 (CH_3-$(CH_2)_4$-CH_3), the Hausdorff dimension of the particle projections remains close to unity ($D_H = 1.124$). Consequently, the measured fringe visibility V remains very high (> 0.96), and the fitted fractional differential order α closely approaches 0.88, matching the theoretical relation $\alpha = 2 - D_H$.

Conversely, as the carbon chain length increases and transitions to complex, highly branched semi-condensed networks and polycyclic aromatic structures (Sample A-105, condensed PAH networks), the Hausdorff dimension increases dramatically to $D_H = 1.915$. This non-Euclidean boundary complexity induces massive local phase decorrelation. The optical interference fringes become severely attenuated, yielding a low visibility ($V = 0.149$ at $\lambda = 532.0$ nm). Fitting the empirical intensity profile $I(\theta)$ to the FOIT equation yields a fractional order parameter $\alpha = 0.086 \pm 0.022$.

Furthermore, a distinct spectral dependence is observed: shorter wavelengths ($\lambda = 532.0$ nm) experience stronger fractional phase perturbation than longer wavelengths ($\lambda = 808.0$ nm). This is attributable to the relative sizing parameter $x = \frac{2\pi a}{\lambda}$; as λ decreases, the optical wave resolves smaller fractal features of the semi-condensed hydrocarbon aggregates, resulting in a more pronounced effective reduction in the fractional order parameter α.

5 Conclusion

This laboratory investigation has successfully formulated, tested, and validated the Fractional-Order Optical Interference Theory (FOIT) as a powerful mathematical paradigm for modeling light propagation through complex atmospheric scattering media. By replacing conventional integer-order spatial derivatives with Caputo fractional derivatives of order $\alpha \in (0, 1]$, the FOIT equation effectively captures non-Gaussian phase perturbations, fringe visibility attenuation, and spatial memory phenomena that classical Mie and scalar wave optics fail to address. The experimental integration of semi-condensed hydrocarbon aerosol synthesized from controlled pyrolysis—evaluated structurally via semi-condensed zigzag notation CH_3-$(CH_2)_n$-CH_3 and polycyclic aromatic frameworks—provided a rigorous empirical benchmark for the theoretical models.

Quantitative analysis of angle-resolved scattering data collected across multiple laser wavelengths ($\lambda = 532.0$ nm, 632.8 nm, 808.0 nm) demonstrated an exact inverse linear coupling between the structural Hausdorff dimension D_H of the scattering aggregates and the empirical fractional differential order parameter, conforming to the relation $\alpha = 2 - D_H$. As the aggregation state transitioned from linear aliphatic chains ($D_H = 1.124, \alpha = 0.878$) to complex condensed aromatic networks ($D_H = 1.915, \alpha = 0.086$), the interference fringe visibility experienced a corresponding non-linear collapse. These findings prove that fractional calculus provides an indispensable mathematical language for remote sensing, atmospheric LiDAR correction algorithms, and particle characterization in turbulent environments. Future research should expand this framework to include vector electromagnetic fractional wave equations and non-stationary transient boundary conditions.

6 References

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